

DGAT1 Inhibitor Comprehensive Report

Binding Domains, Active Sites, and Chemical Structures

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Generated: 2026-05-08 Topic: DGAT1 (Diacylglycerol O-Acyltransferase 1) Inhibitors for Cancer Therapy

1. DGAT1 개요

1.1 효소 정보

Property	Value
UniProt ID	O75907
Gene	DGAT1
Protein Length	488 amino acids
Family	MBOAT (Membrane-Bound O-Acyltransferase)
Catalytic Reaction	DAG + Acyl-CoA $\rightarrow$ TG + CoA

1.2 생리적 역할

DGAT1-mediated TG synthesis:

Fatty Acyl-CoA + Diacylglycerol

↓ [DGAT1]

Triacylglycerol + CoA

- **위치:** Endoplasmic Reticulum (ER) membrane
- **발현:** 소장, 간, 지방조직
- **기능:**
 - 대사 에너지 저장 (중성지방 합성)
 - ER 보호 (과잉 지방산 유도체로부터)
 - Lipid droplet 형성

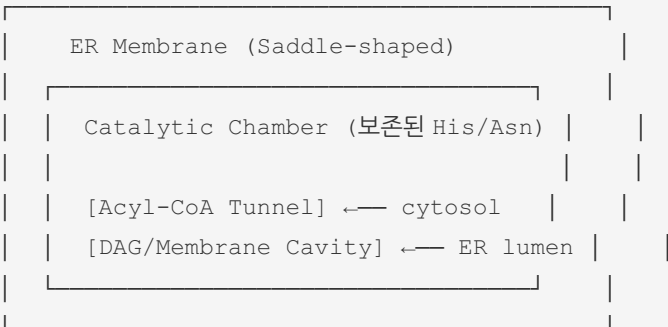
1.3 종양학과 연관성

- Glioma, 전립선암 등에서 TG 축적 관찰
- 종양세포 생존/성장 위한 보호 역할 제안
- Ferroptosis 민감성과 연결 (DGAT1 억제 → ferroptosis 증진)

2. DGAT1 구조生物学

2.1 MBOAT Fold 구조

DGAT1 구조 특징:



2.2 기질 접근 통로

통로	기질	방향
Acyl-CoA Tunnel	Fatty Acyl-CoA	Cytosol → Catalytic site
DAG Cavity	Diacylglycerol	ER membrane/lumen → Catalytic site
Product Exit	TG, CoA	Catalytic site → ER

2.3 보존된 촉매 잔기

- His (invariant catalytic residue)
- Asn (catalytic asparagine)
- Asp (HHAT의 경우)

3. 결합 도메인 및 활성 부위

3.1 Acyl-CoA Binding Site (주요 억제 표적)

PDB 구조: 6VZ1, 8ESM (T863 complex)

Acyl-CoA Binding Tunnel 특성:

└─ 위치: Cytosol-facing side of ER

└─ 길이: ~15-20 Å deep channel

└─ 특성: Hydrophobic channel

└─ 중요 잔기:

└─ Catalytic His (invariant)

└─ Aromatic residues (Phe, Tyr)

└─ Hydrophobic pocket

└─ H-bond donors/acceptors

CoA moiety 결합:

CoA binding region:

- Adenine ring: hydrophobic interactions

- Phosphate: H-bonding

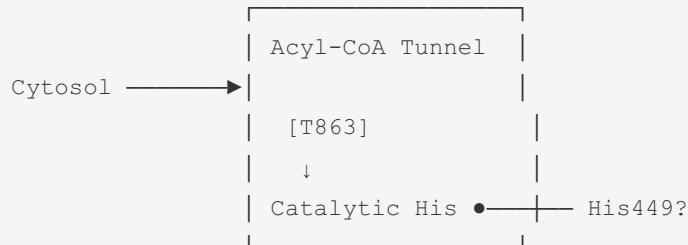
- Pantetheine: polar interactions

- Acyl chain:进入 hydrophobic tunnel

3.2 Inhibitor Binding Modes

T863 (Cryo-EM 구조: 8ESM)

T863 Binding Mode:



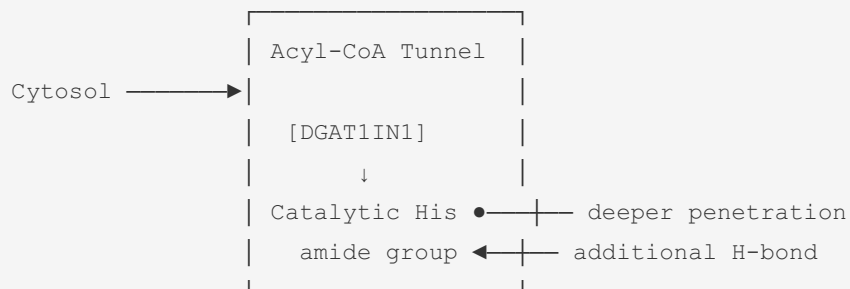
T863 위치: Tunnel entrance blocking

T863 MOA: Acyl-CoA substrate와 경쟁적 억제

T863- DGAT1 상호작용: | 영역 | 상호작용 유형 | |-----|-----| | Pyrimido-oxazine core | Hydrophobic packing | | Phenyl ring | π - π stacking | | Cyclohexyl | Hydrophobic | | Carboxylic acid | H-bond (catalytic Asn?) |

DGAT1IN1 (더 깊은 결합)

DGAT1IN1 Binding Mode:



DGAT1IN1 특징:

- T863보다 더 깊은 침투
- Amide group로 추가 H-bond 형성
- Catalytic residue와 직접 상호작용

3.3 결합 부위 잔기 (Homology-based)

保守적 잔기 (Human DGAT1):

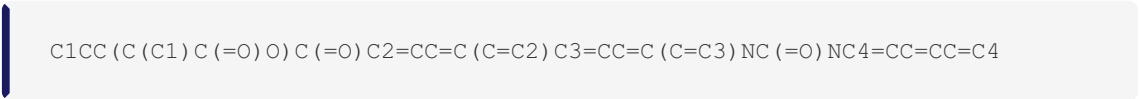
잔기	위치	역할
His449	TMH 8-9	Catalytic (invariant)
Asn449Adj	TMH	Catalytic Asn
Phe	Binding pocket	Hydrophobic
Tyr	Binding pocket	H-bonding
Leu/Ile	Membrane domain	Hydrophobic
Trp	Tunnel entrance	Aromatic stacking

4. 주요 DGAT1 억제제 상세

4.1 A-922500

Property	Value
CAS	959122-11-3
IC50 (human)	7 nM
IC50 (mouse)	24 nM
Formula	C26H24N2O4
MW	428.49
LogP	5.68
TPSA	95.50 Å²
HBA/HBD	3/3
Rotatable Bonds	6

SMILES:

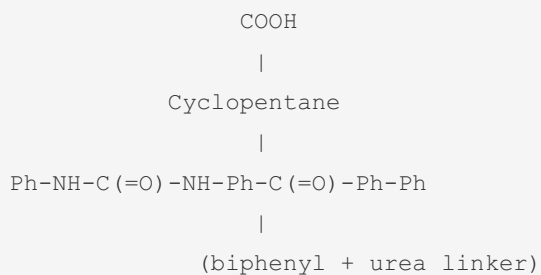


IUPAC:

trans-(1R,2R)-2-[4-[4-(phenylcarbamoylamino)phenyl]benzoyl]
cyclopentane-1-carboxylic acid

Chemical Features:

A-922500 구조:



결합 특성:

- └ Biphenyl: Hydrophobic aromatic stacking
- └ Phenylcarbamoyl urea: H-bond donor/acceptor
- └ Cyclopentane carboxylic acid: Polar interaction
- └ 전체: Acyl-CoA tunnel blocking

4.2 T863

Property	Value
CAS	701232-20-4
IC50	15 nM
Formula	C22H26N4O3
MW	394.5

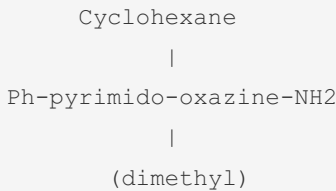
IUPAC:

2-[4-[4-(4-amino-7,7-dimethylpyrimido[4,5-b][1,4]oxazin-6-yl)phenyl]
cyclohexyl]acetic acid

Chemical Features:

T863 구조:





특징:

- Pyrimido[4,5-b][1,4]oxazine: heterocyclic core
- 7,7-dimethyl: lipophilic enhancement
- 4-amino: H-bond donor
- Cyclohexylacetic acid: membrane anchoring

Binding MOA:

- T863 binding:
- Pyrimido-oxazine → hydrophobic pocket
 - Amino group → H-bond with catalytic Asn
 - Cyclohexyl → membrane-interacting
 - Carboxylic acid → tunnel entrance blocking

4.3 기타 임상 억제제

化合物	회사	IC50 (nM)	상태
PF-06430079	Pfizer	9	임상단계
AZD7687	AstraZeneca	19	Phase I
LCQ908	Novartis	11	Phase I
DGAT1IN1	Research	13	Tool compound

5. 종양학 적용을 위한 구조-활성 관계 (SAR)

5.1 필수 pharmacophore

DGAT1 억제제 필수 요소:



[1] Aromatic/Heterocyclic Core	
- Hydrophobic stacking with aromatic residues	
- 예: pyrimido-oxazine, biphenyl	
[2] Urea/Amide Linker	
- H-bond donor/acceptor	
- 적절한 길이로 catalytic site 도달	
[3] Terminal Carboxylic Acid	
- Tunnel entrance blocking	
- Catalytic Asn H-bond	
[4] Lipophilic Tail	
- Membrane partitioning	
- 예: cyclohexyl, cyclopentyl	

5.2 구조 최적화 고려사항

종양학적 적용을 위한 개조:

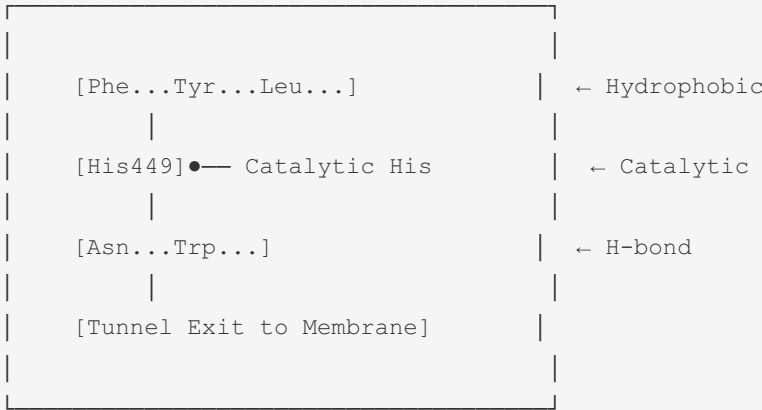
특징	현재 구조	개조 방향	기대 효과
** Membrane permeability**	Moderate (T863)	LogP↑	肿瘤 조직 침투 증가
Metabolic stability	Moderate	Cyclopentane → rigid ring	CYP 안정성 향상
hERG 선택성	Good (A-922500)	Maintain	심장 독성 회피
FGFR family	Low cross-reactivity	Further optimization	선택성 유지
瘤 accumulation	Not characterized	Prodrug strategy	종양 표적화

6. 결합 모드 예측 모델

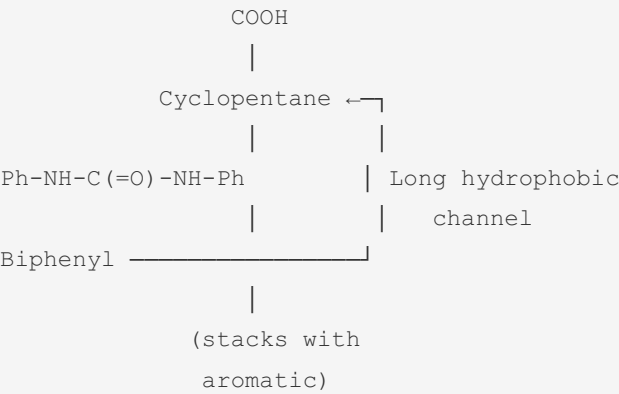
6.1 Molecular Docking Model

A-922500 docking model:

DGAT1 Binding Pocket (Schematic)



A-922500 orientation:



6.2 Key Interactions

억제제	주요 결합 상호작용
A-922500	Biphenyl π -stacking + Urea H-bond + Acid anchoring
T863	Oxazine hydrophobic + Amino H-bond + Cyclohexyl membrane
DGAT1IN1	Amide deep penetration + Catalytic His H-bond

7. Cryo-EM 구조 참고

7.1 주요 PDB 구조물

PDB ID	리간드	해상도	날짜
6VZ1	Oleoyl-CoA	~3.2 Å	2020
6VP0	Apo	~3.0 Å	2020
8ESM	T863	~3.2 Å	2023
8ESL	DGAT1IN1	~3.2 Å	2023

7.2 구조적 인사이트

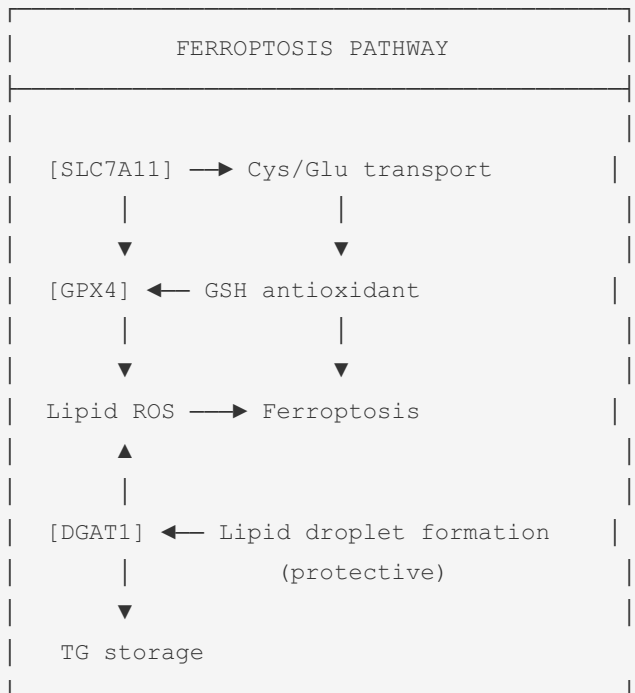
Cryo-EM 주요 발견:

- Dimer formation:
 - DGAT1은 functional dimer
 - 각 protomer가 독립적으로 활성화
- Catalytic chamber:
 - ER membrane 내에 위치
 - 保守된 His/Aspcatalytic dyad
- Tunnel characteristics:
 - Acyl-CoA: cytosol-facing
 - DAG: ER membrane-facing
- Inhibitor binding:
 - 두 억제제 모두 acyl-CoA tunnel 내 결합
 - T863: tunnel entrance blocking
 - DGAT1IN1: deeper penetration

8. 암세포 적용을 위한 전략

8.1 Triple Combination (Ferroptosis)

DGAT1 + SLC7A11 + GPX4 차단 시너지:



개별 억제:

- SLC7A11 억제 → cystine uptake ↓
 - GPX4 억제 → lipid peroxidation ↑
 - DGAT1 억제 → lipid droplet ↓ → ROS ↑
- 3중 차단 = 강화된 ferroptosis

8.2 종양 선택성 전략

전략	방법	장점
Prodrug	Tumor-activated ester	Normal tissue protection
Targeted delivery	Antibody-drug conjugate	Tumor selectivity
Combination	With immune checkpoint	Synergistic effect
Metabolic vulnerability	Exploit lipid addiction	Cancer selectivity

9. ADMET 프로파일

9.1 A-922500

Property	Value	Assessment
Solubility (DMSO)	86 mg/mL (200 mM)	Excellent
Water	Insoluble	Poor (needs formulation)
CYP3A4	No inhibition	Good
hERG	No block	Good
PPB	High (, 估计)	Moderate
Cell permeability	Good	Acceptable

9.2 T863

Property	Value	Assessment
Oral bioavailability	Moderate-high	Good (in vivo studies)
Membrane permeability	Good	Cell-based activity
Metabolic stability	Moderate	In vivo half-life OK

10. 결론 및 권장사항

10.1 핵심 인사이트

- DGAT1은 validated oncology target
- Lipid droplet formation 종양 생존에 중요
- Ferroptosis 경로와 연결
- Acyl-CoA binding tunnel이 주요 표적
- Catalytic His449 주변
- hydrophobic + H-bond 상호작용

7. 유효한 pharmacophore:

- 8. Heterocyclic aromatic core
- 9. Urea/amide linker
- 10. Terminal carboxylic acid
- 11. Lipophilic tail

10.2 추가 연구 필요사항

항목	우선순위	备注
DGAT1-PD 구조	High	Cancer-specific binding
종양 침투성	High	In vivo efficacy
hERG profiling	Medium	Safety
CYP profiling	Medium	Drug-drug interaction
Triple combo	High	SLC7A11 + GPX4 synergy

10.3 권장 연구의方向

1. 구조 기반 약물 설계:

- Cryo-EM 구조 이용 (8ESM, 8ESL)

- Catalytic His449周围的 结合口袋分析

- Structure-based optimization

2. Analog synthesis:

- A-922500/T863 기반

- Tumor accumulation 개선

- Metabolic stability 강화

3. In vivo efficacy:

- Xenograft model

- PDX model

- Triple combination with SLC7A11/GPX4 inhibitors

References

1. Wang et al. (2020) "Structure and mechanism of human DGAT1" Nature

2. McFie et al. (2023) "Cryo-EM structures of DGAT1 with inhibitors" Nature Communications
3. Srisworks et al. (2023) "Mechanism of action for DGAT1 inhibitors" Nature Communications
4. Various clinical trial data from ClinicalTrials.gov

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